

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : DISHGUARD 72

Other means of identification : Not applicable.

Recommended use : Machine Warewashing Detergent

Product dilution information : 0.75 % - 1.0 %
0.75 - 1.0%

Company : ECOLAB PTE LTD
21 Gul Lane
Singapore 629416
65-6505-6868

Emergency telephone number : +1-651-222-5352 (United States)

Issuing date : 26.12.2017

Section: 2. HAZARDS IDENTIFICATION
GHS Classification
Product AS SOLD

Acute toxicity (Oral) : Category 4

Skin corrosion/irritation : Category 1

Serious eye damage/eye irritation : Category 1

Carcinogenicity : Category 2

Acute aquatic toxicity : Category 1

Product AT USE DILUTION

Not a hazardous substance or mixture.

GHS Label element
Product AS SOLD

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Harmful if swallowed.
Causes severe skin burns and eye damage.
Suspected of causing cancer.
Very toxic to aquatic life.

Precautionary Statements : **Prevention:**
Obtain special instructions before use. Do not breathe dust/fume/gas/mist/vapours/spray. Wash skin thoroughly after handling. Avoid release to the environment. Wear protective gloves/ protective clothing/ eye protection/ face protection. Do not eat, drink or smoke when using this product. Do not handle until all safety precautions have been read and understood.
Response:

SAFETY DATA SHEET

DISHGUARD 72

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/doctor. IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell. Rinse mouth. Do NOT induce vomiting. Wash contaminated clothing before reuse. IF exposed or concerned: Get medical advice/attention. Collect spillage.

Storage:

Store locked up.

Product AT USE DILUTION

Precautionary Statements : **Prevention:**
Wash hands thoroughly after handling.
Response:
Get medical advice/ attention if you feel unwell.
Storage:
Store in accordance with local regulations.

Product AS SOLD

Other hazards : None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Alcohols, C13-15, branched and linear, ethoxylated	157627-86-6	>= 30 - < 60
Didecyl Dimethyl Ammonium Chloride	7173-51-5	>= 10 - < 30
2,2'-iminodiethanol	111-42-2	>= 0.1 - < 1
Fattyalcohol alkanolamides	61791-31-9	>= 10 - < 30
Isopropanol	67-63-0	>= 1 - < 5

Product AT USE DILUTION

No hazardous ingredients

Section: 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention immediately.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal

SAFETY DATA SHEET

DISHGUARD 72

protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Section: 5. FIREFIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : None known.

Specific hazards during firefighting : Fire Hazard
Keep away from heat and sources of ignition.
Flash back possible over considerable distance.
Exposure to decomposition products may be a hazard to health.
Beware of vapours accumulating to form explosive concentrations.
Vapours can accumulate in low areas.

Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
nitrogen oxides (NOx)

Special protective equipment for firefighters : Use personal protective equipment.

Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Remove all sources of ignition. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

SAFETY DATA SHEET

DISHGUARD 72

Methods and materials for containment and cleaning up : Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Section: 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Keep away from fire, sparks and heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours). Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing.

Conditions for safe storage : Keep away from heat and sources of ignition. Keep away from oxidizing agents. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 40 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
Isopropanol	67-63-0	PEL (long term)	400 ppm 983 mg/m ³	SG PEL
		PEL (short term)	500 ppm 1,230 mg/m ³	SG PEL
2,2'-iminodiethanol	111-42-2	PEL (long term)	0.46 ppm 2 mg/m ³	SG PEL

SAFETY DATA SHEET

DISHGUARD 72

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Impervious gloves
Laminate film
Nitrile
Unsupported neoprene
PVC
Natural rubber
Neoprene/natural rubber blend
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.
Wash face, hands and any exposed skin thoroughly after handling.
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Colour	: colourless	colourless
Odour	: odourless	slight
pH	: 3.5 - 5.5, (100 %)	3.5 - 4.5
Flash point	: 45 °C closed cup, Does not sustain combustion.	
Odour Threshold	: no data available	

SAFETY DATA SHEET

DISHGUARD 72

Melting point/freezing point	: no data available
Initial boiling point and boiling range	: no data available
Evaporation rate	: no data available
Flammability (solid, gas)	: no data available
Upper explosion limit	: no data available
Lower explosion limit	: no data available
Vapour pressure	: no data available
Relative vapour density	: no data available
Relative density	: 0.981 - 1.001
Water solubility	: soluble
Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: 75.817 mm ² /s (40 °C)
Explosive properties	: no data available
Oxidizing properties	: The substance or mixture is not classified as oxidizing.
Molecular weight	: no data available
VOC	: no data available

Section: 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, flames and sparks.
Incompatible materials	: None known.
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NO _x)

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns.

SAFETY DATA SHEET

DISHGUARD 72

Ingestion : Harmful if swallowed. Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Suspected of causing cancer.

Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Toxicity

Product AS SOLD Product

Acute oral toxicity : Acute toxicity estimate : 1,690 mg/kg

Acute inhalation toxicity : 4 h Acute toxicity estimate : 0.47 mg/l
Test atmosphere: dust/mist

Acute dermal toxicity : no data available

Skin corrosion/irritation : no data available

Serious eye damage/eye irritation : no data available

Respiratory or skin sensitization : no data available

Carcinogenicity : no data available

Reproductive effects : no data available

Germ cell mutagenicity : no data available

SAFETY DATA SHEET

DISHGUARD 72

Teratogenicity : no data available

STOT - single exposure : no data available

STOT - repeated exposure : no data available

Aspiration toxicity : no data available

Components

Acute dermal toxicity : Alcohols, C13-15, branched and linear, ethoxylated
LD50 rat: > 2,000 mg/kg

Didecyl Dimethyl Ammonium Chloride
LD50 rabbit: 2,930 mg/kg

2,2'-iminodiethanol
LD50 rabbit: 8,180 mg/kg

Isopropanol
LD50 rabbit: 12,870 mg/kg

Section: 12. ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects : Very toxic to aquatic life.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other
aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Didecyl Dimethyl Ammonium Chloride
96 h LC50 Fish: 1 mg/l

Fattyalcohol alkanolamides
96 h LC50: 0.47 mg/l

Isopropanol
96 h LC50 Pimephales promelas (fathead minnow): 9,640 mg/l

Components

Toxicity to daphnia and other
aquatic invertebrates : Alcohols, C13-15, branched and linear, ethoxylated
48 h EC50 Daphnia magna (Water flea): 0.317 mg/l

2,2'-iminodiethanol
48 h EC50 Daphnia: 65.5 mg/l

Fattyalcohol alkanolamides
48 h EC50: 0.38 mg/l

Isopropanol
LC50 Daphnia magna (Water flea): > 10,000 mg/l

Persistence and degradability

SAFETY DATA SHEET

DISHGUARD 72

Product AS SOLD

Biodegradable/Eliminated from aquatic environment

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

The product should not be allowed to enter drains, water courses or the soil.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

UN number : 1760
Description of the goods : CORROSIVE LIQUID, N.O.S.
(Alkyl ammonium chloride)
Class : 8
Packing group : III
Hazchem Code : -
Environmentally hazardous : Yes

Air transport (IATA)

UN number : 1760
Description of the goods : Corrosive liquid, n.o.s.
(Alkyl ammonium chloride)
Class : 8
Packing group : III

SAFETY DATA SHEET

DISHGUARD 72

Environmentally hazardous : No

Sea transport (IMDG/IMO)

UN number : 1760
Description of the goods : CORROSIVE LIQUID, N.O.S.
(Alkyl ammonium chloride)
Class : 8
Packing group : III
Marine pollutant : Yes

Special precautions for user : None

Section: 15. REGULATORY INFORMATION

National Regulations

Fire Safety (Petroleum and Flammable Materials) Regulations

Isopropanol

The components of this product are reported in the following inventories:

United States TSCA Inventory :

On TSCA Inventory

Canadian Domestic Substances List (DSL) :

This product contains one or several components listed in the Canadian NDSL.

Australia. Industrial Chemical (Notification and Assessment) Act :

not determined

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :

On the inventory, or in compliance with the inventory

Japan. ENCS - Existing and New Chemical Substances Inventory :

not determined

Korea. Korean Existing Chemicals Inventory (KECI) :

not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

not determined

China Inventory of Existing Chemical Substances :

On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :

On the inventory, or in compliance with the inventory

Section: 16. OTHER INFORMATION

Issuing date : 26.12.2017
Version : 1.0
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

SAFETY DATA SHEET

DISHGUARD 72

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.